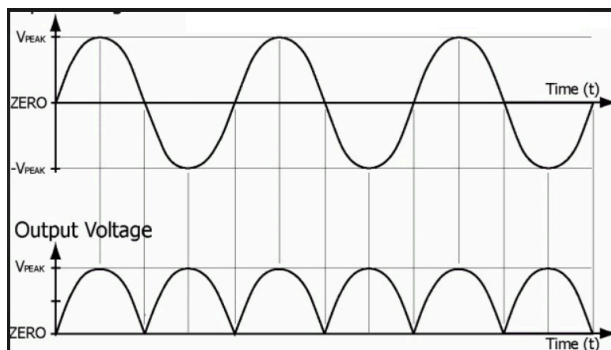




NSN MEMORIAL SENIOR
SECONDARY SCHOOL
PHYSICS INVESTIGATORY PROJECT
2025-2025
FULL WAVE RECTIFIER WITH ARDUINO
DSO

Page 1



SUBMITTED BY
S.SAHANA
CLASS: XII -B



N.S.N. MEMORIAL SENIOR SECONDARY SCHOOL CHENNAI

BONAFIDE CERTIFICATE

*Certified to be the bonafide project work in
done by of Standard
during the academic year 20__ - 20__*

Register No:

Signature of the Principal

School Seal

*Submitted for the Senior Secondary Practical Examination
held on at N.S.N. Memorial Senior Secondary
School, Chennai - 64.*

INTERNAL EXAMINER

EXTERNAL EXAMINER

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Name of the student: SAHANA.S

Register number:

Signature:

OBJECTIVE

The objective of this project is to design and construct a Full-Wave Rectifier circuit to convert alternating current (AC) into direct current (DC), and to study its working, output characteristics, and ripple reduction using a filter capacitor. An Arduino-based digital oscilloscope setup is integrated as an add-on feature to observe and analyze the input AC and rectified DC waveforms in real-time. The project demonstrates the process of rectification, filtering, and safe voltage measurement using a potential divider, resistor, and Zener diode protection network.

INTRODUCTION

In electronic circuits, converting alternating current (AC) into direct current (DC) is one of the most essential processes, as most electronic devices operate on DC voltage. A Full-Wave Rectifier is a circuit that converts both halves of the AC input signal into DC output, making it more efficient than a half-wave rectifier. It uses a pair of diodes (or a bridge of four diodes) to allow current to flow during both positive and negative halves of the AC cycle.

To further improve the output, a **filter capacitor** is connected across the load resistor to reduce voltage fluctuations and produce a smoother DC output. This process is known as **filtering** or **smoothing**.

In this project, a **Full-Wave Bridge Rectifier** is designed using **IN4007 diodes**, a **step-down transformer**, and a **filter capacitor**. To observe and analyze the waveform of the rectified output, an **Arduino-based digital oscilloscope** is used as an add-on feature. The Arduino measures the output voltage through a **potential divider** and a **Zener diode protection circuit**, ensuring safe input levels for the analog pin.

This integration helps visualize the transformation of AC to DC, study the effect of filtering, and understand the practical behavior of rectifier circuits in a low cost way.

ABOUT THE TOPIC

Hence, rectification, the process of converting AC into DC, is one of the most important applications in power electronics.

A Full-Wave Rectifier is a circuit that utilizes both halves of the AC waveform to produce a continuous flow of current in one direction. Compared to a half-wave rectifier, it offers greater efficiency, higher average output voltage, and less ripple. The most practical and widely used form is the Bridge Rectifier, which employs four diodes arranged in a diamond shape. This configuration eliminates the need for a center-tapped transformer and provides full-wave rectification using simple components.

During operation, the diodes conduct in pairs depending on the polarity of the AC cycle, ensuring that the load current always flows in the same direction. This produces a pulsating DC output, which, although not perfectly smooth, can be further refined using filters. The output of a full-wave rectifier is therefore the basic form of DC used in most power supply units, chargers, and converters.

In this project, the rectifier circuit is coupled with an Arduino-based oscilloscope to analyze the output waveform. The Arduino reads the voltage variations through its analog input pin, which is protected by a voltage divider and a Zener diode network. This allows the microcontroller to safely measure voltage between 0-5V.



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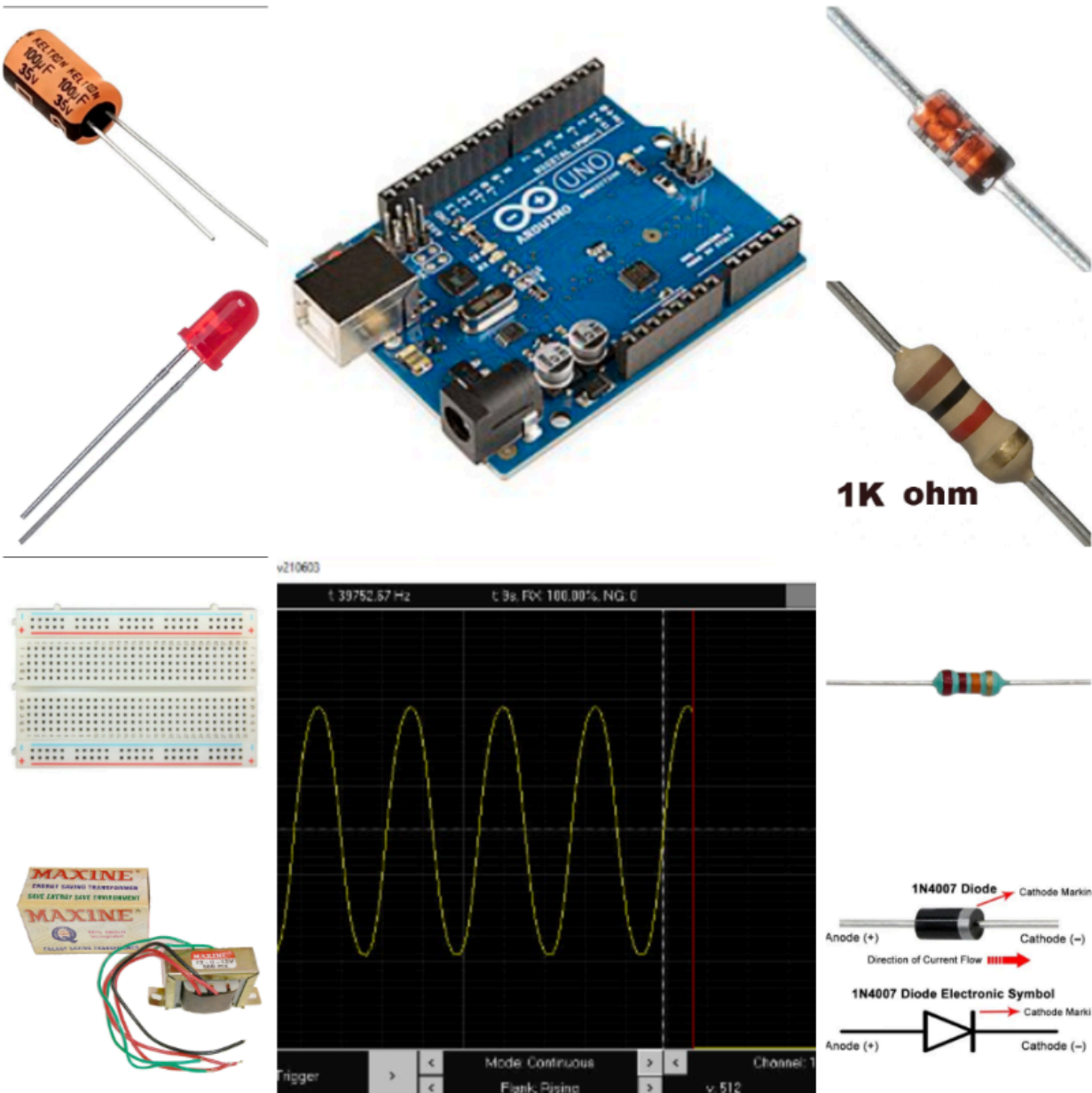
EXPERIMENT

AIM:

To construct a **Full-Wave Bridge Rectifier** and analyze its output DC waveform using an **Arduino-based oscilloscope**, including ripple measurement and verification using a multimeter.

APPARATUS REQUIRED:

SNO	COMPONENT	RATING
1	IN4007 Diodes	1 A, 1000 V
2	Step-down Transformer	12V 500mA
3	Load Resistor	1k 1W
4	Capacitor	100 mF 25V
5	LED	red
6	Arduino uno	Any
7	Resistors(voltage divider)	10k and 4k
8	Zener diode	5.1V
9	Bread board	any
10	Multimeter	any
11	Zaida digital pc scope	DSO
12	Jumper wires	any



FORMULA:

1. RMS of AC input

$$V_{\text{rms}} = \frac{V_m}{\sqrt{2}}$$

2. Voltage Divider for Arduino input

$$V_{\text{out}} = V_{\text{in}} \times \frac{R_2}{R_1 + R_2}$$

3. Ripple Voltage Approximation

$$V_{\text{ripple}} \approx \frac{I_{\text{load}}}{f_r C}$$

4. DC Voltage Approximation

$$V_{\text{DC}} \approx V_{\text{peak}} - \frac{V_{\text{ripple}}}{2}$$

COMPONENT TESTING: (Before Experiment)

1. Diodes (IN4007)

- Set multimeter to diode mode.
- Forward bias: reading $\approx 0.6\text{--}0.7\text{ V}$.
- Reverse bias: reading “OL” (no conduction).

2. Resistors

- Use ohmmeter to verify resistance values.

3. Capacitor

- Use multimeter (capacitance mode) to check value; ensure no short.

4. LED

- Connect series resistor ($\sim 330\ \Omega$) to 5 V supply; LED should glow.

5. Zener diode

- Can test in diode mode: forward conducts at $\sim 0.7\text{ V}$; reverse shows OL until breakdown ($\sim 5.1\text{ V}$)

THEORY:

1. AC INPUT AND STEP DOWN TRANSFORMER

The mains AC voltage is stepped down to ~12 V using a transformer for safe operation. This reduced AC is then fed to the bridge rectifier.

2 . BRIDGE RECTIFICATION:

Four IN4007 diodes form a bridge configuration:

- Positive half-cycle: Diodes D1 and D2 conduct, allowing current to flow through the load in one direction.
- Negative half-cycle: Diodes D3 and D4 conduct, again directing current through the load in the same direction.

This results in a pulsating DC voltage across the load resistor.

3.FILTERING:

A 100 μF capacitor connected across the load smooths the pulsating DC by charging at voltage peaks and discharging at troughs. This reduces ripple voltage, making the output more steady. A LED connected across the load glows steadily when the voltage is sufficiently smooth.

4. VOLTAGE DIVIDER AND ARDUINO INPUT:

The rectified DC is scaled down using a voltage divider before feeding it to the Arduino's analog input (A0):

- R1 (top resistor): 10 k Ω

- R2 (bottom resistor): 4.7 k Ω

This ensures the voltage at A0 stays below 5 V, within Arduino's safe analog input range.

5. ZENER DIODE PROTECTION: (Safety Circuit):

A 5.1 V Zener diode is connected in reverse bias across the Arduino input and ground.

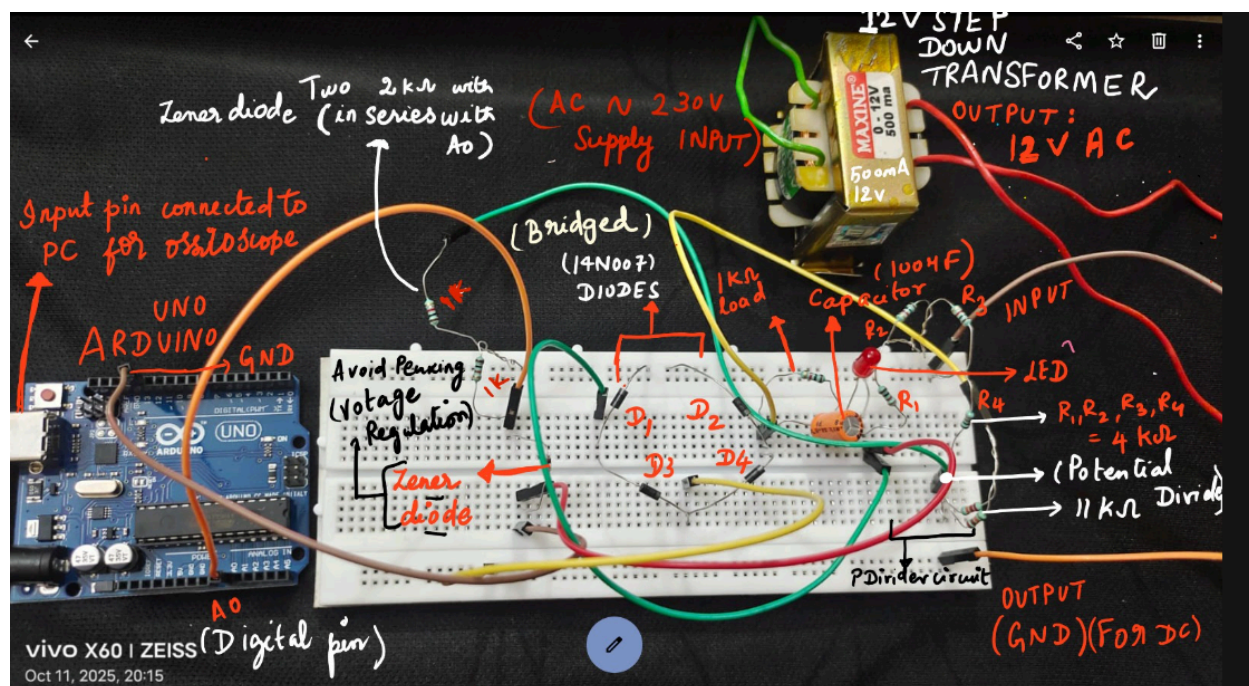
- Purpose: Protects the Arduino from voltage spikes that might exceed 5 V.
- Working: Under normal conditions, the divider output (~3.8–4 V) is below the Zener voltage, so the diode does not conduct. If the voltage rises above 5.1 V, the Zener clamps it, preventing damage to the microcontroller.

6. WAVE FORM OBSERVATION:

The Arduino reads the scaled voltage waveform through A0 and sends the data to a PC, where software like Zaida DSO displays the waveform digitally. This allows real-time observation of:

- AC-to-DC conversion
- Ripple voltage
- Effectiveness of the filter capacitor
- Steadiness of DC output

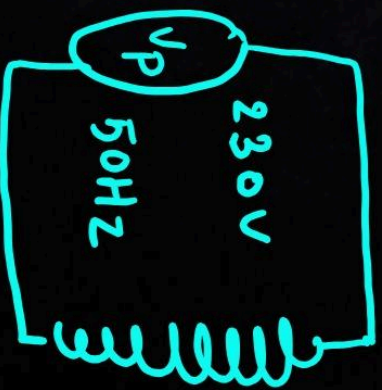
CIRCUIT AND BREAD BOARD CONNECTIONS:



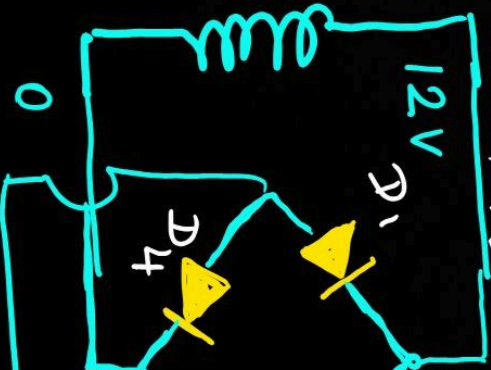
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CIRCUIT

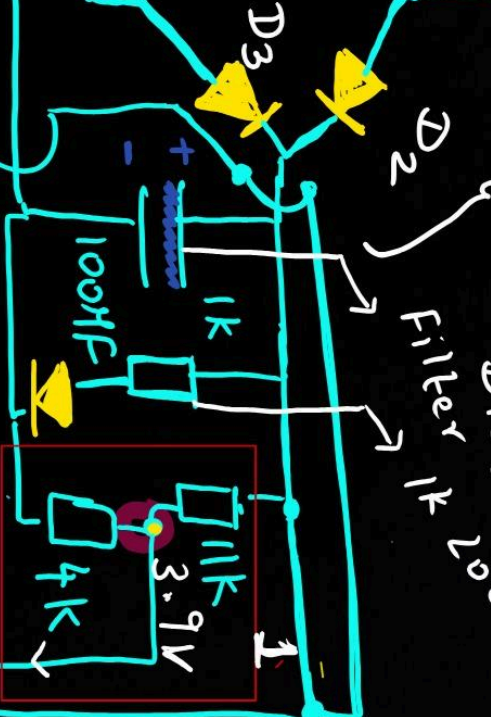
Full Wave Rectifier:



Transformer
(Step down)



Bridged
Diodes
Filter 1k 20a



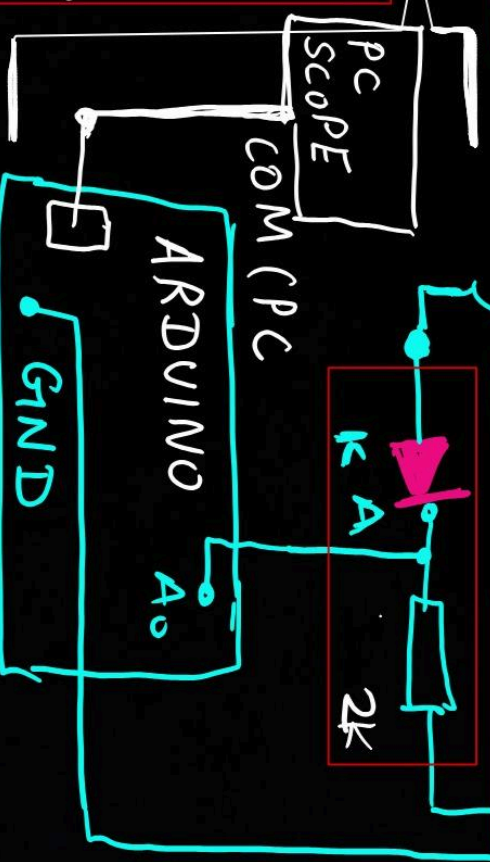
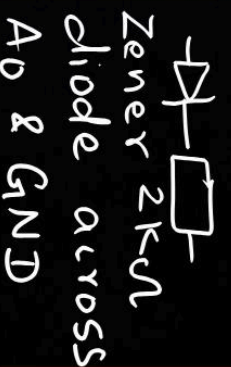
DSO Basic



1] Potential
Divider
Circuit

$$V_{out} = \frac{11k}{11k + 4k} \times 3.54V$$

2] Anti Spike
mech for
Arduino.



CIRCUIT CONNECTIONS:

1. Bridge Rectifier

- Connect 4 IN4007 diodes in a bridge:
Two AC terminals → Transformer secondary
- terminal → Load resistor → LED (optional)
- terminal → Load resistor → common ground

2. Filter

- Capacitor across load resistor (parallel).

3. Arduino Input

- Connect load output → voltage divider → series resistor → A0.
- Voltage divider: R1 (top) = 10 k Ω , R2 (bottom) = 4.7 k Ω (parallel to filter capacitor)
- Zener diode: cathode → A0, anode → common ground.

4. Common Ground

- Arduino GND connected to bridge rectifier negative and voltage divider bottom. then upload arduino code and connect with ZAIDA digital oscilloscope.

PROCEDURE:

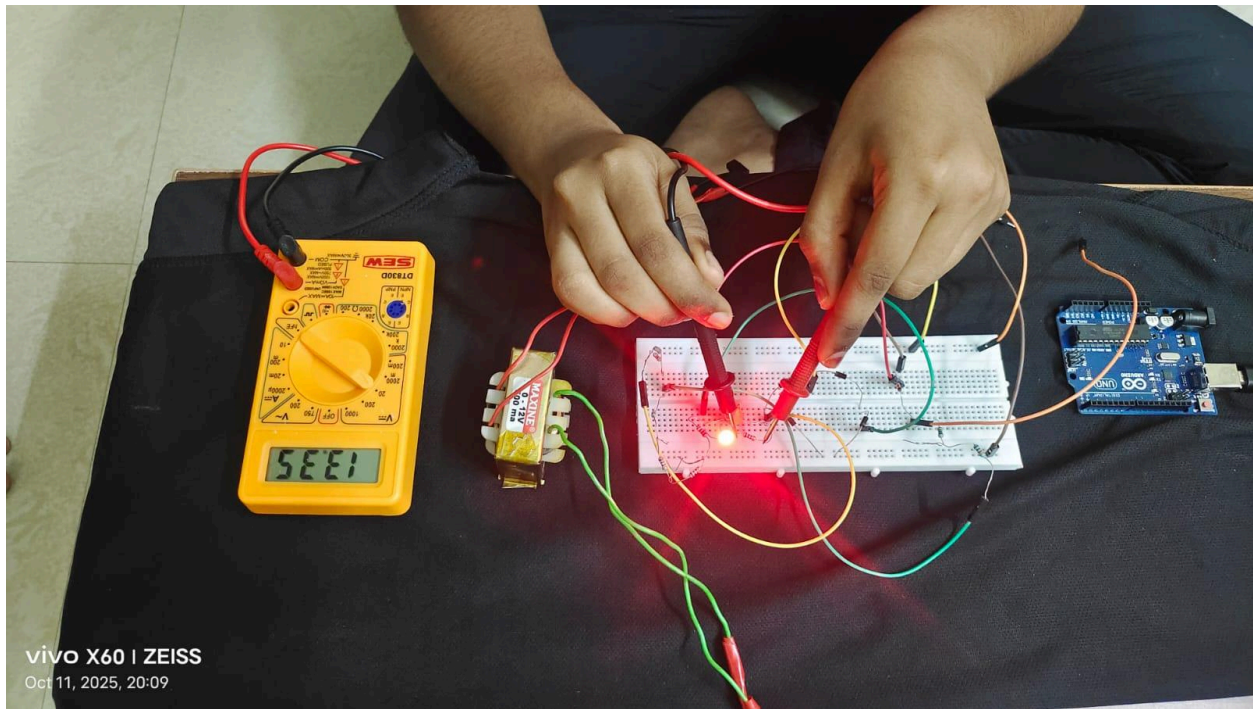
1. Test all components as described above.
2. Connect the bridge rectifier on the breadboard.
3. Connect step-down transformer secondary to bridge AC input.
4. Connect load resistor + capacitor + LED across rectifier output. Observe LED glow.
5. Build voltage divider + Zener + series resistor for Arduino input. Connect to A0.
6. Upload Arduino DSO sketch. Open Zaida DSO on PC.
7. Observe the rectified waveform on PC screen. Note DC level, ripple, and shape.
8. Measure rectified DC voltage across load resistor using a multimeter.
9. Record observations (DC voltage, ripple amplitude).
10. Compare readings with theoretical calculation

OBSERVATION (USING MULTIMETER) AND OSSILOSCOPE:

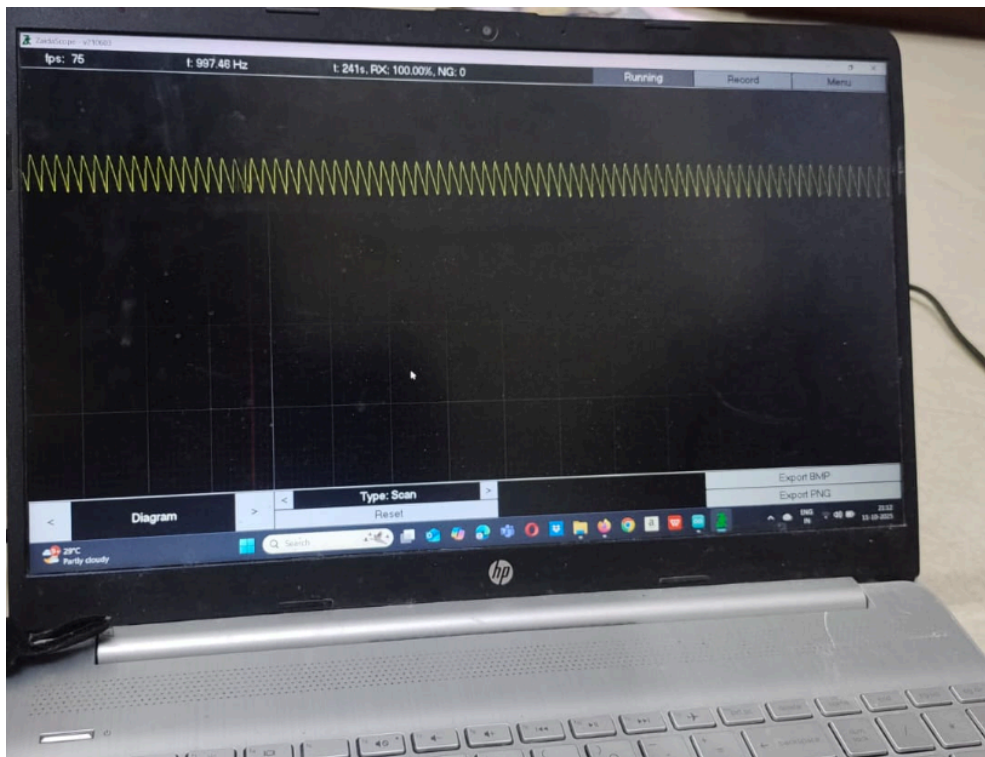
PARAMATER	THEORY	PRACTICAL	REMARKS
INPUT(STEP DOWN VOLTAGE)	12V with 16.9V(V_o) 33.2V(V_{pp})	13.2V with 18.66(V_o) 37.33(V_{pp})	From multimeter in ac mode
VOLATAGE AFTER DIODE DROP	1.4V	1.4V	—
AVG DC VALUE AFTER RECTIFYING	15.57V (16.9V-1.4V)	13.3V	From multimeter in DC mode
RIPPLE VOLATAGE	1.3V	cant measure with cheap DMM	Across capacitor
POTENTIAL DRAWN FROM DIVIDER	3.9V	3.86V	Between ground and 4k resistor.
LED STATUS	GLOWS	GLOWS	BRIGHT
(Arduino) ADC	782-783	782-783	Confirms that the voltage is below 5V

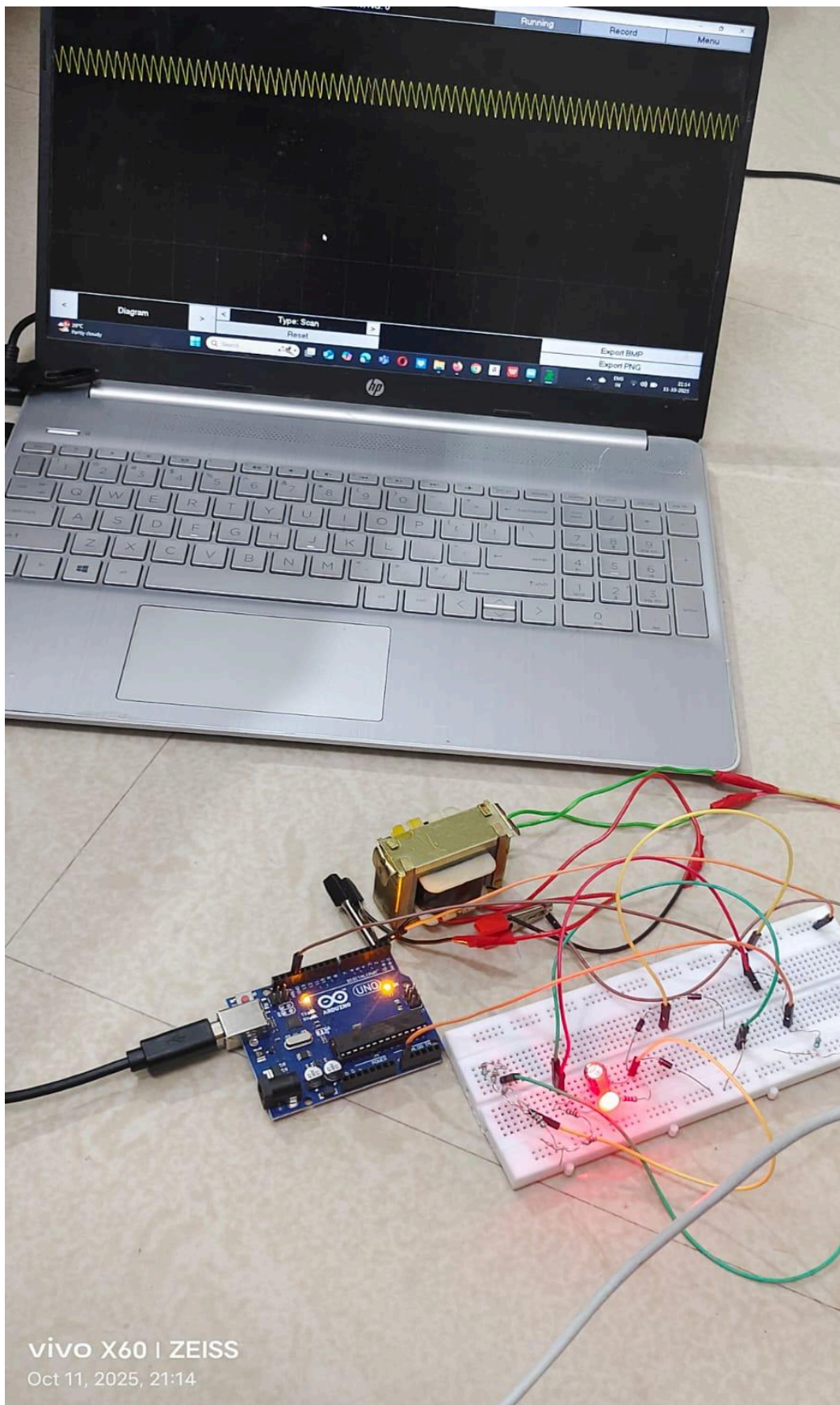
- (1) Reading in multimeter 13.3V rectified dc
*below image)
(2)Oscilloscope output without filter)

1)



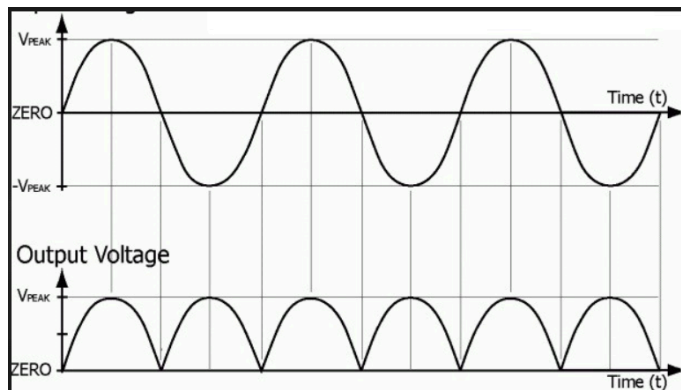
2)



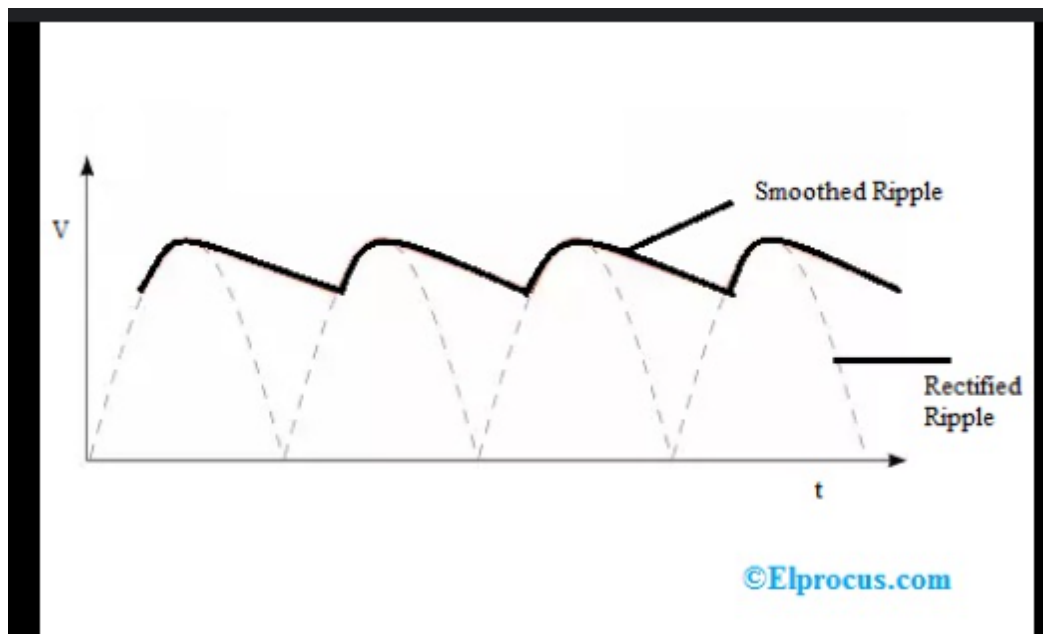


THEORITICAL RECTIFIED OUTPUT GRAPH:

1) AC TO PULSATING DC (WITHOUT FILTER)



2) AC TO RIPPLE DC WITH FILTER



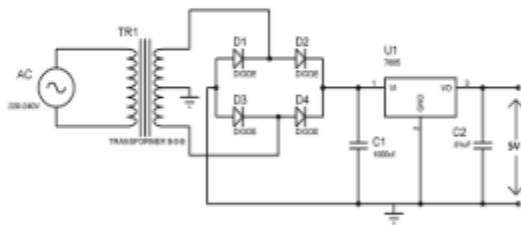
PRECAUTIONS:

1. Test all components with a multimeter before use.
2. Connect diodes and capacitor with correct polarity.
3. Use a voltage divider to keep Arduino input ≤ 5 V.
4. Connect Zener diode in reverse bias for protection.
5. Avoid short circuits and ensure proper grounding.
6. Use transformer to step down AC safely.

APPLICATIONS

1. Phone Chargers – Converts AC mains to DC to charge batteries.
2. Laptop Adapters – Provides steady DC output for powering laptops.
3. LED Lighting Systems – Powers LED bulbs and strips from AC mains.
4. TV and Audio Equipment – Supplies DC voltage to internal circuits.
5. Battery Charging Circuits – For rechargeable batteries in toys, gadgets, and UPS systems.
6. DC Motors in Appliances – Small DC motors in printers, fans, and other devices.

Mobile Phone Charger Circuit



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